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2/1/2026

Proof of Central Limit Theorem

Let $X_1, X_2, \ldots, X_n$ be an iid random sample from any PDF. Let $S_n = \sum_{i=1}^{n} X_i$

Now let $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$

Note: Z_n has mean 0 and variance σ^2

See that $S_n - n\mu = \sum_{i=1}^{n} (X_i - \mu) = \sum_{i=1}^{n} Y_i$

We want to show that $\lim_{n\to\infty} P(Z_n \leq x) = \phi(x)$ where $\phi(x)$ is the CDF of the Normal Distribution.

Now let $M_{Z_n}(t)$ be the moment generating function of Z_n

$$M_{Z_n}(t) = E\left(e^{tZ_n}\right) = E\left[\exp\left(\frac{t\sum Y_i}{\sigma\sqrt{n}}\right)\right]$$

Taking the sum out of the exponent gives

$$\prod_{i=1}^{n} E\left[\exp\left(\frac{tY_i}{\sigma\sqrt{n}}\right)\right]$$

Note: $M_{Y_i}(s) = E\left(e^{sY_i}\right)$

For each Y_i, $E\left[\exp\left(\frac{tY_i}{\sigma\sqrt{n}}\right)\right] = M_{Y_i}(s)$ where $s = \frac{t}{\sigma\sqrt{n}}$

Now $\prod_{i=1}^{n} E\left[\exp\left(\frac{tY_i}{\sigma\sqrt{n}}\right)\right] = \prod_{i=1}^{n} M_{Y_i}(s) = \left[M_Y(s)\right]^n$

Since each Y_i is identically distributed, they have the same MGF, call it $M_Y(s)$

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Now find the taylor series expansion of $M_y(s)$ around 0

$$M_y(s) \approx M_y(0) + s M_y'(0) + \frac{s^2}{2!} M_y''(0) + \dots$$

Note: $M_y(0) = 1$, $M_y'(0) = E(y) = 0$

$M_y''(0) = E(y^2) = \sigma^2$

Recall: $y = x - \mu$

so $E(y^2) = E[(x-\mu)^2] = \sigma^2$

Now $M_y(s) = 1 + \frac{\sigma^2}{2!} s^2$

$= 1 + \frac{\sigma^2}{2} \cdot \frac{t^2}{\sigma^2 n} = 1 + \frac{t^2}{2n}$

we don't care about higher order terms.

Next term: $s^3/3! \, M_y'''(0)$

$= \frac{t^3/\sigma^3 n^{3/2}}{3!} M_y'''(0)$

$\lim_{n\to\infty} 1/n^{3/2} = 0$

Now $[M_y(s)]^n = \left[1 + \frac{t^2}{2n}\right]^n \longrightarrow n \ln\left(1 + \frac{t^2}{2n}\right)$

Let $x_n = \frac{t^2}{2n}$ then $n = \frac{t^2}{2} \cdot \frac{1}{x_n}$ so $\frac{t^2}{2} \frac{\ln(1+x_n)}{x_n}$

Then $\lim_{n\to\infty} x_n = 0$ so $\lim_{x_n \to 0} \frac{\ln(1+x_n)}{x_n} = 1$

So $\lim_{n\to\infty} \ln(M_y(s)) = \frac{t^2}{2}$

Take $e^{\square}$ of each side and get $e^{t^2/2}$

This is the moment generating function of the normal distribution $\square$

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